

Does the Spatial Architecture of a Tumor Predict Patient Outcome?

Testing whether *where* immune cells sit – not just *how many* – carries prognostic signal, with a spatial graph neural network

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The problem. Whether a cancer patient does well often depends not just on *which* cells are in their tumor but on *how those cells are spatially arranged* – in particular, whether immune cells *infiltrate* the tumor (often favorable) or are *excluded* from it (often unfavorable). Two tumors can have the *identical* cell composition yet opposite outcomes because of this arrangement. Standard analyses summarize a tumor by its cell *proportions* and therefore cannot see this. This project asks, rigorously: **does the spatial arrangement of the tumor microenvironment add prognostic signal beyond composition?**

1 Background and Significance

The tumor microenvironment (TME) – the ecosystem of tumor, immune, and stromal cells – is a central determinant of cancer progression and response to therapy. A growing body of pathology shows that the *spatial organization* of this ecosystem is prognostic: “immune-inflamed” tumors with cytotoxic T cells in contact with tumor cells tend to respond better than “immune-excluded” tumors where those same T cells are trapped at the margin. New technologies (imaging mass cytometry, CODEX, spatial transcriptomics) now measure dozens of markers per cell *while preserving each cell’s location*, making it possible to quantify TME architecture at single-cell resolution.

Why it matters	Consequence
Outcome depends on cell <i>arrangement</i>	Two tumors with the same composition can have opposite prognoses.
Composition summaries discard geometry	Proportion-based models are blind to infiltration vs. exclusion.
Spatial single-cell data now exist	We can finally test the arrangement hypothesis directly.

Table 1: Spatial arrangement of the TME is widely believed prognostic; testing whether it adds signal beyond composition is the question we address.

2 The Unmet Need

Gap 1 – Composition hides geometry. Summarizing a tumor by its cell proportions cannot distinguish an infiltrated tumor from an excluded one with the same cell counts.

Gap 2 – Most spatial models *assume* arrangement matters. Many TME outcome models report accuracy without isolating whether the gain comes from spatial structure or merely from composition – so the central biological claim is rarely tested.

We close Gap 2 with a controlled, falsifiable comparison (Fig. 1): a model that reads spatial arrangement vs. strong composition-only models, plus an ablation that destroys the arrangement.

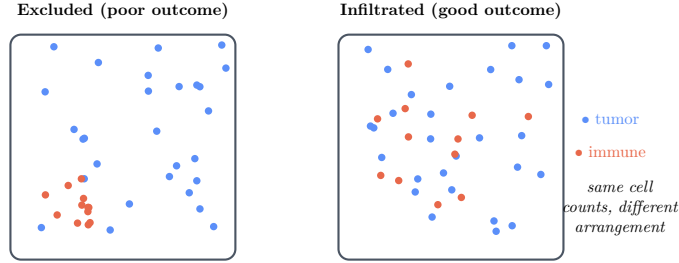


Figure 1: Identical composition, opposite arrangement. Composition-only models see the two tumors as the same; a spatial model reads the tumor–immune contact structure that distinguishes them.

3 Research Aims

Aims

Aim 1. Represent each patient’s tumor as a spatial cell graph and predict a clinical outcome (tumor grade and/or survival) with a graph neural network.

Aim 2. Test *honestly* whether spatial arrangement adds signal *beyond composition* – compare against tuned composition-only baselines (XGBoost/MLP on cell proportions) and a graph-shuffle ablation.

Aim 3. Interpret which spatial cell-neighbourhoods drive prognosis (candidate prognostic niches), grounding predictions in TME biology.

Guiding questions. (Q1) Can outcome be predicted from TME spatial structure? (Q2) Does it beat a composition-only model, and is the gain *causally* due to arrangement? (Q3) Which spatial niches are prognostic?

4 Data

5 Proposed Methodology

Each patient’s tumor is a graph: nodes are cells (features = cell-type / marker profile), edges connect spatially neighbouring cells. We pose outcome prediction as **graph classification**.

5.1 Proposed model: a spatial GNN

A GraphSAGE/GIN encoder pools each cell’s neighbourhood over L hops, then a global readout summarizes the whole tumor:

$$h_v^{(l)} = \sigma\left(W^{(l)}\left[h_v^{(l-1)} \parallel \text{AGG}_{u \in \mathcal{N}(v)} h_u^{(l-1)}\right]\right), \quad \hat{y} = \text{softmax}\left(\text{MLP}\left(\text{POOL}_v h_v^{(L)}\right)\right). \quad (1)$$

Resource	Content	Role
Jackson-Fischer breast IMC	~700 patients, 37 markers, single-cell coords; tumor grade + survival	Real cohort (primary)
METABRIC IMC	483 tumors, clinical outcomes	External validation
Controlled synthetic cohort	infiltration vs. exclusion, composition matched	Mechanism proof / ablation

Table 2: All inputs are public, de-identified. Coordinates enter only through the spatial graph, never as features.

The readout depends on neighbourhood structure (e.g. tumor-immune contacts), so the model can represent arrangement, not just composition.

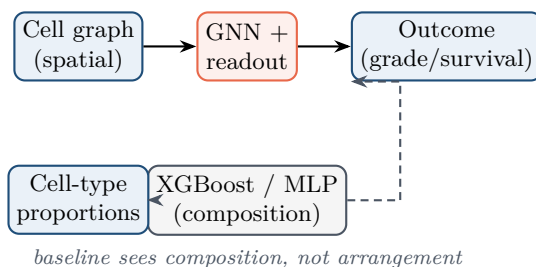


Figure 2: Pipeline. The GNN (orange) reads the spatial cell graph; the composition baselines (gray) see only aggregated cell proportions.

5.2 Baselines and rigorous evaluation

Model	Sees arrangement?	Purpose
Spatial GNN (ours)	yes	prognosis from TME architecture
XGBoost / MLP	no (composition only)	strong arrangement-blind baselines
Logistic regression	no (composition only)	simple control

Table 3: Baselines receive cell-type proportions + marker summaries; only the GNN sees the spatial graph. Any gap is the value of arrangement.

Patient-level splits (no patient in two folds), 5 seeds. **Metrics:** accuracy, macro-F1, AUROC for grade; concordance index for survival. **Falsification:** a *graph-shuffle* / *empty-graph ablation* – if the GNN’s gain is real, randomizing the spatial edges must collapse it onto the composition baseline.

6 Interpretation and Impact

We will identify the spatial cell-neighbourhoods (e.g. tumor-immune contact motifs) that drive high-confidence predictions, yielding *candidate prognostic niches* that can be cross-checked against known TME biology. If spatial arrangement adds signal beyond composition, that supports spatial

architecture as a biomarker; if it does not, that is itself an important, honestly-reported negative result. Either way the question – not a new algorithm – is the contribution.

7 Expected Outcomes

Outcome	Target
Controlled proof	On composition-matched synthetic tumors, GNN $\gg$ composition baselines; ablation confirms arrangement is the signal.
Real cohort	Quantify whether the spatial GNN adds over composition on Jackson-Fischer IMC (reported as-is).
Interpretation	A shortlist of candidate prognostic spatial niches.
Honesty clause	Composition is expected to carry real signal on real tumors; we report the <i>added</i> value of arrangement honestly, not an inflated headline.

Table 4: Success is a rigorous, interpretable answer to whether TME arrangement adds prognostic signal beyond composition.

8 Eight-Week Timeline

Weeks	Milestones
1–2	IMC data pipeline (cells $\rightarrow$ per-patient spatial graphs); controlled synthetic cohort.
3–4	Spatial GNN + composition baselines (XGBoost/MLP/LogReg); shared trainer.
5	Patient-level experiments, 5 seeds, grade + survival metrics.
6	Graph-shuffle ablation; composition-vs-arrangement decomposition.
7	Interpretation: prognostic niche extraction; external validation (METABRIC).
8	Figures, tables, manuscript; reproducibility package.

Table 5: Schedule for the eight-week SRI cohort.

9 Reproducibility and Ethics

Implementation uses open tools (PyTorch Geometric; scikit-learn/XGBoost). All code, configs, seeds, and a data manifest will be released. No human subjects are involved; all inputs are public, de-identified tumor tissue data. Predictions are research-only and hypothesis-generating and are not clinical guidance.

Selected References

1. Jackson, Fischer, et al. *The single-cell pathology landscape of breast cancer*. Nature, 2020.
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3. Wu, et al. *Graph neural networks for spatial proteomics predict patient outcomes from the tumor microenvironment*. (cell-graph prognosis).
4. Hamilton, Ying, Leskovec. *Inductive Representation Learning on Large Graphs (GraphSAGE)*. NeurIPS, 2017.
5. Xu, et al. *How Powerful are Graph Neural Networks? (GIN)*. ICLR, 2019.